Research Data Camp: Data Publishing and Repositories

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Agenda

1. What is data publishing?
2. Why publish data?
3. FAIR Data Principles
4. Top considerations for collecting and sharing data
5. CU Scholar example
6. Questions and wrap-up

1. What is data publishing?

Working definition

- Making research data and metadata/documentation publicly available (or with appropriate access controls) via a formal web-based repository/database
- Preferably in adherence with FAIR data principles and/or other standards for data, metadata, and repository quality

Related terms

- Data sharing
- Data curation
- Data archiving
- Data preservation

2. Why publish data?

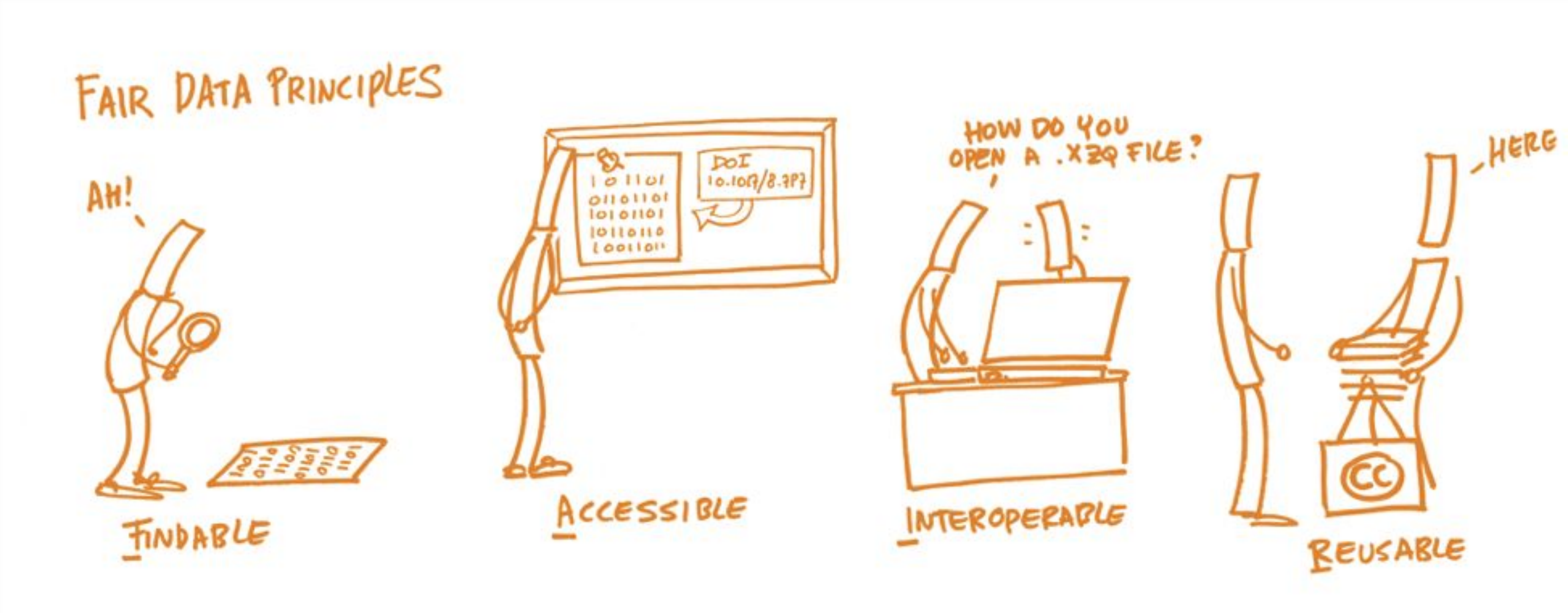
Why publish data?

1. Scientific and public good
    a. Advance scientific innovation
    b. Address reproducibility
2. Journal/publisher requirements
    a. “Data availability statements”
    b. FAIR-aligned repositories with citations via persistent IDs
3. Funder requirements
    a. Part of NSF data management plans since 2011
    b. Part of NIH data management and sharing policy since 2023

Thinking Ahead

3. FAIR Data Principles

Introduction to FAIR data principles (Wilkinson et al., 2016)

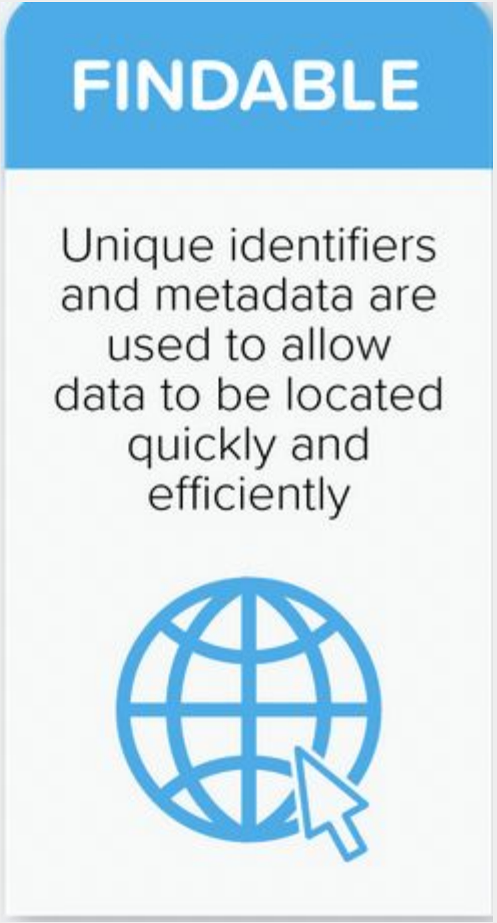


OpenAire: "How to Make Your Data FAIR"

Image by Patrick Hochstenbach, CC0 1.0

Findable (F)

- Apply a globally unique and persistent identifier
- Describe your data in a data repository



https://www.go-fair.org/fair-principles/

Accessible (A)

- Consider what will be shared, and share via a open, free, and universally implementable protocol
- Metadata are valuable and accessible, even when the data are no longer available

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Interoperable (I)

- Use:
  - Open formats
  - Consistent vocabulary
  - Common metadata standards

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Reusable (R)

- Origin, context, history, and who to credit/cite are all crucial for data reusability
- Consider permitted use and apply the appropriate license

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4. Top considerations for collecting and sharing data

Top Considerations for You # 1

Have clear documentation and a data management plan from square one.

- Think ahead about repositories and requirements for a finished project
- Document throughout the process/project: how data was created/gathered/used/etc.
- R (Reusable) in FAIR is very hard to achieve just at the end of the project; important to think about from the beginning

Top Considerations for You # 1

Before you start collecting data, think about:

- How much of your data will you/can you share?
- How and where will you share your data?
- When will you share your data?
- With whom will you share your data?

Top Considerations for You # 2:

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Select a FAIR-aligned data repository

- CU Scholar
  - FAIR-aligned public access repository for CU Boulder affiliated researchers (i.e., have an IdentiKey)
  - Has CoreTrustSeal certification
  - Review and curation of all data sets
  - DataCite DOIs registered for all data sets
  - Public access to large data sets via Globus and PetaLibrary
  - Free to deposit up to 500 GB per data set for CU Boulder affiliated researchers
  - Over 2300 datasets published in CU Scholar to date

Top Considerations for You # 2:

Select a FAIR-aligned data repository

- General repositories (e.g., Dryad, Dataverse, Zenodo)
  - Open to anyone to deposit
  - Minimal review/curation of deposits
  - Typically provide DataCite DOIs and usage metrics
  - Size limits and/or additional fees for large data
- Domain repositories:
  - Re3data repository registry
  - Level of review/curation varies
  - Ability to deposit varies
  - May be recommended/required by funders/publishers

Top Considerations for You # 3:

Consider copyright and licensing of your data set

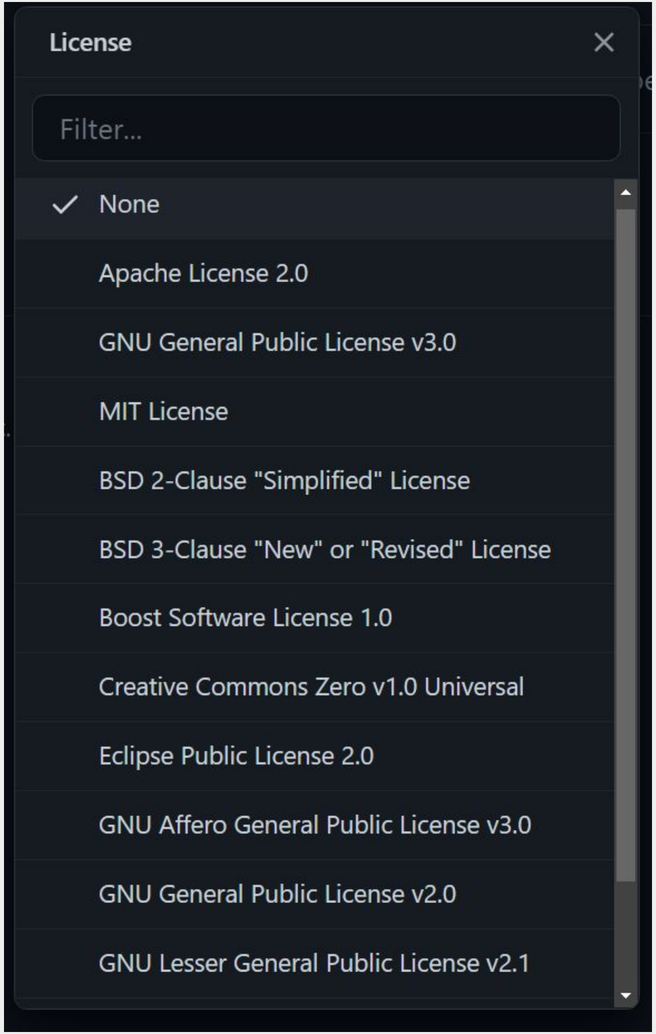
- The license that is selected facilitates sharing and reuse of the data set
    - [Creative Commons]()
    - CC BY: Creative Commons Attributions License
    - CC 0: When an owner wishes to waive their copyright and/or database rights
    - Public Domain mark (PDM): It is used to mark works that are in the public domain, and for which there are no known copyright or database restrictions.

Top Considerations for You # 3:

Consider copyright and licensing of associated software/code

- Many licenses available for software/code

GitHub



5. CU Scholar

Example Data Set in CU Scholar

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Data Set

Thermal Structure of the Martian Upper Mesosphere/Lower Thermosphere from MAVEN/IUVS Stellar Occultations [Data] Public Deposited

Analytics

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Citations Citations: ▾

Citeable URL: https://scholar.colorado.edu/concern/datasets/h702q775d

Abstract

The MAVEN/IUVS stellar occultation dataset is publicly available at the NASA planetary data system's atmosphere node. But it consists of only the nightside events and limited dayside observations. We have reprocessed the mission-wide dataset from March 2015 to January 2022 using an improved stray light removal algorithm to retrieve the dayside events as well, thereby expanding the usable stellar occultation dataset and enabling the study of diurnal thermal structure of the upper mesosphere/lower thermosphere (~80-160 km).

We have provided this reprocessed dataset and the retrieved data products according to the executed campaigns, along with the output of a global-mean 1-D numerical model.

Creator

Gupta, Sumedha

Academic Affiliation

Laboratory for Atmospheric & Space Physics

Last Modified

2022-08-12

Resource Type

Data Set

Rights Statement

In Copyright

DOI

https://doi.org/10.25810/z1wy-cq62

Language

English [eng]

Citation

Gupta, S. (2022). Thermal Structure of the Martian Upper Mesosphere/Lower Thermosphere from MAVEN/IUVS Stellar Occultations [Data] [Data set]. University of Colorado Boulder . https://doi.org/10.25810/Z1WY-CQ62

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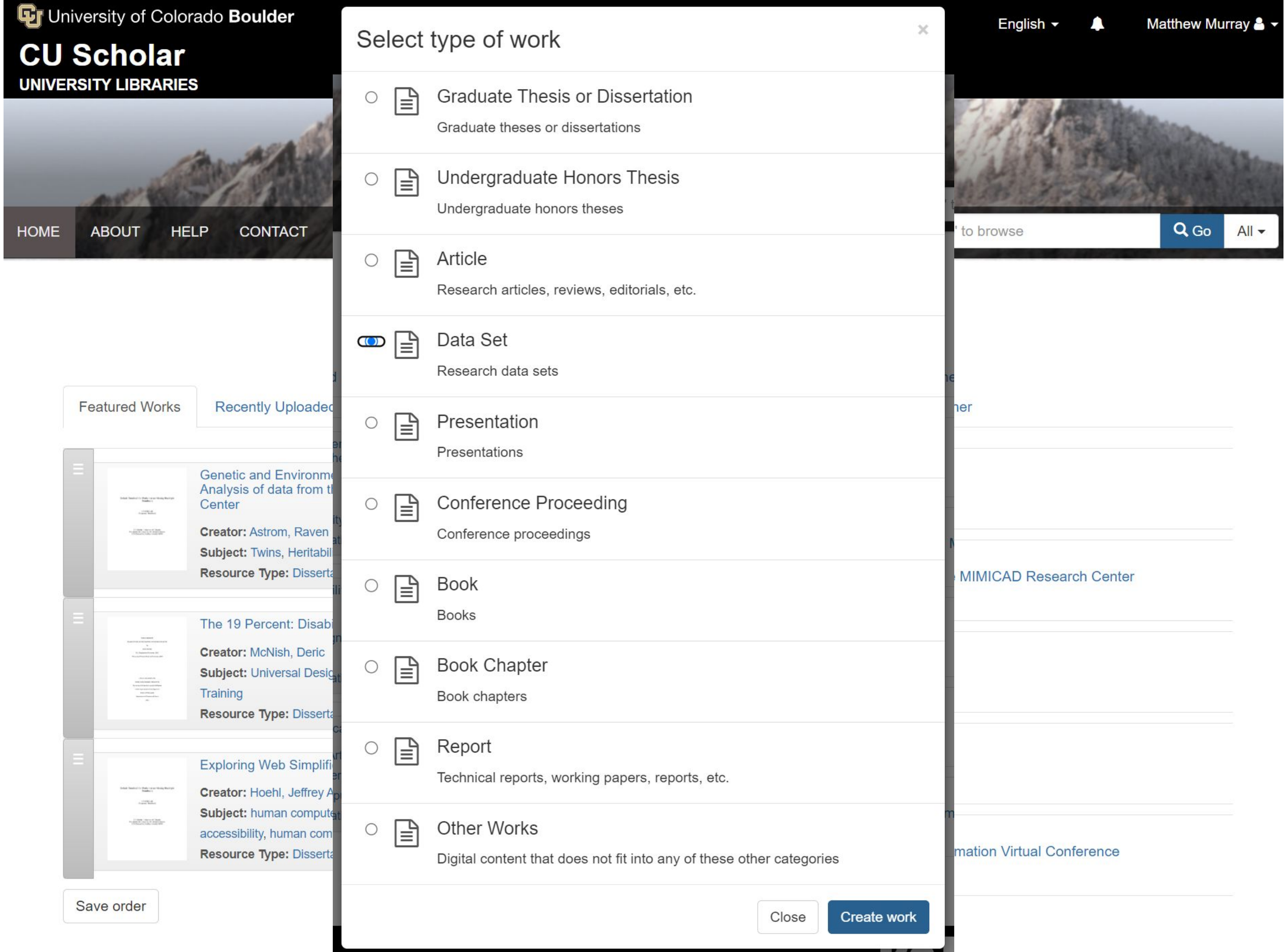
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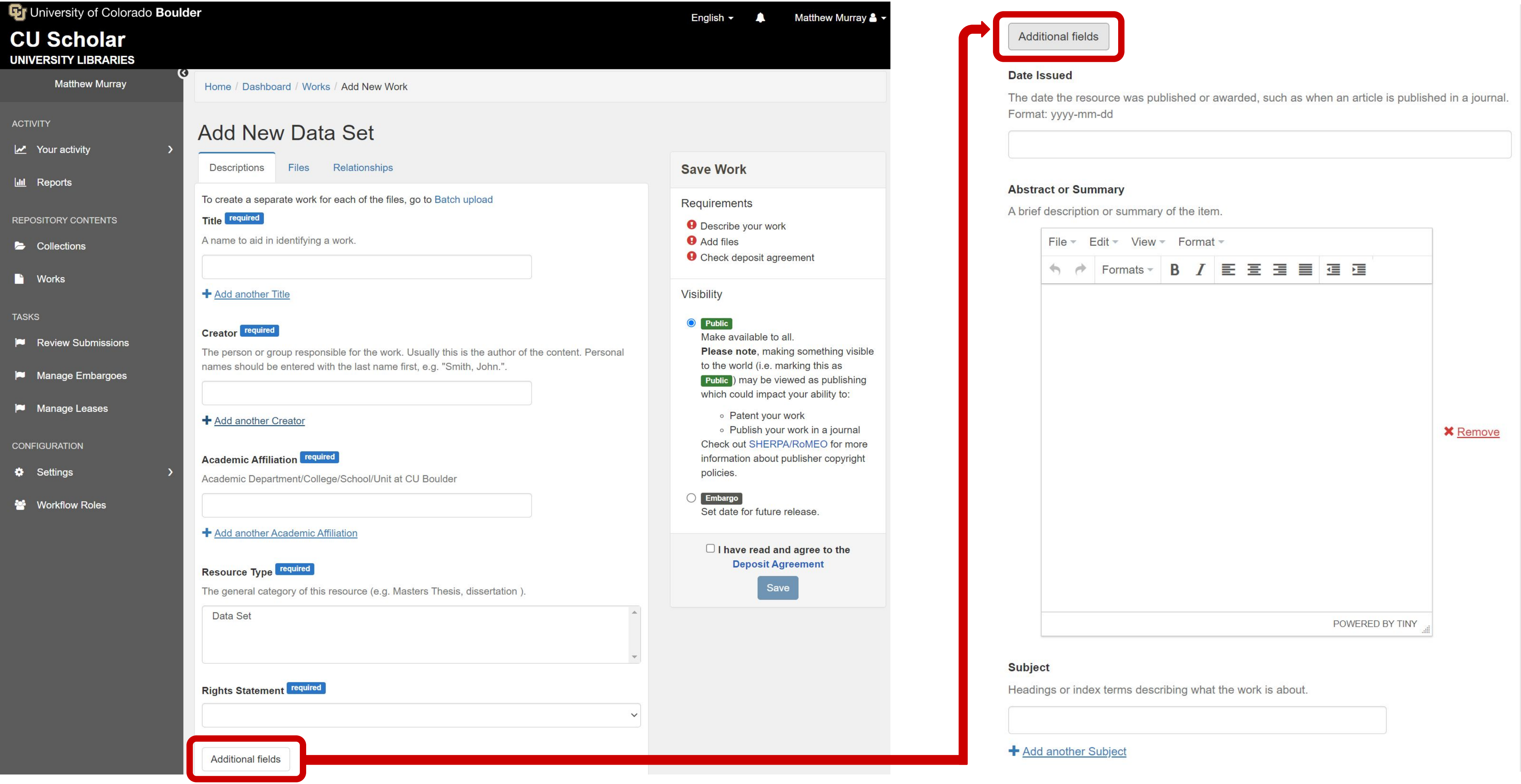
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